

Experiment 6: Demonstration of Apriori algorithm on transaction dataset to find association rules

Aim

To implement the Apriori algorithm on a transaction dataset to discover frequent itemsets and generate association rules based on support, confidence, and lift.

Code

```
import pandas as pd
import numpy as np

# -----
# STEP 1: Load Dataset
# -----
# Dataset has transactions row-wise (each row = items purchased)
dataset = pd.read_csv("Market_Basket_Optimisation.csv", header=None)

# Convert dataset into list of transactions
transactions = []
for i in range(len(dataset)):
    transaction = [str(dataset.values[i, j]) for j in range(20) if str(dataset.values[i, j]) != 'nan']
    transactions.append(transaction)

# -----
# STEP 2: Convert to One-Hot Encoding
# -----
from mlxtend.preprocessing import TransactionEncoder

te = TransactionEncoder()
te_data = te.fit(transactions).transform(transactions)
df = pd.DataFrame(te_data, columns=te.columns_)

# -----
# STEP 3: Apply Apriori Algorithm
# -----
from mlxtend.frequent_patterns import apriori

frequent_itemsets = apriori(df, min_support=0.01, use_colnames=True)

print("Frequent Itemsets:\n", frequent_itemsets.head())

# -----
# STEP 4: Generate Association Rules
# -----
from mlxtend.frequent_patterns import association_rules
```

```

rules = association_rules(frequent_itemsets, metric="lift", min_threshold=1)

# Sort rules by lift
rules = rules.sort_values(by="lift", ascending=False)

print("\nTop Association Rules:\n")
print(rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']].head(10))

```

Output

Frequent Itemsets:

	support	itemsets
0	0.020397	(almonds)
1	0.033329	(avocado)
2	0.010799	(barbecue sauce)
3	0.014265	(black tea)
4	0.011465	(body spray)

Top Association Rules:

	antecedents	consequents	support \
214	(herb & pepper)	(ground beef)	0.015998
215	(ground beef)	(herb & pepper)	0.015998
386	(ground beef)	(spaghetti, mineral water)	0.017064
383	(spaghetti, mineral water)	(ground beef)	0.017064
397	(olive oil)	(spaghetti, mineral water)	0.010265
396	(spaghetti, mineral water)	(olive oil)	0.010265
193	(tomatoes)	(frozen vegetables)	0.016131
192	(frozen vegetables)	(tomatoes)	0.016131
189	(shrimp)	(frozen vegetables)	0.016664
188	(frozen vegetables)	(shrimp)	0.016664

	confidence	lift
214	0.323450	3.291994
215	0.162822	3.291994
386	0.173677	2.907928

383	0.285714	2.907928
397	0.155870	2.609786
396	0.171875	2.609786
193	0.235867	2.474464
192	0.169231	2.474464
189	0.233209	2.446574
188	0.174825	2.446574

Result

The Apriori algorithm was successfully applied to the transaction dataset to identify frequent itemsets based on minimum support. Association rules were generated using metrics such as confidence and lift. The strongest rules were identified by sorting based on lift, revealing meaningful relationships between items that are frequently purchased together.